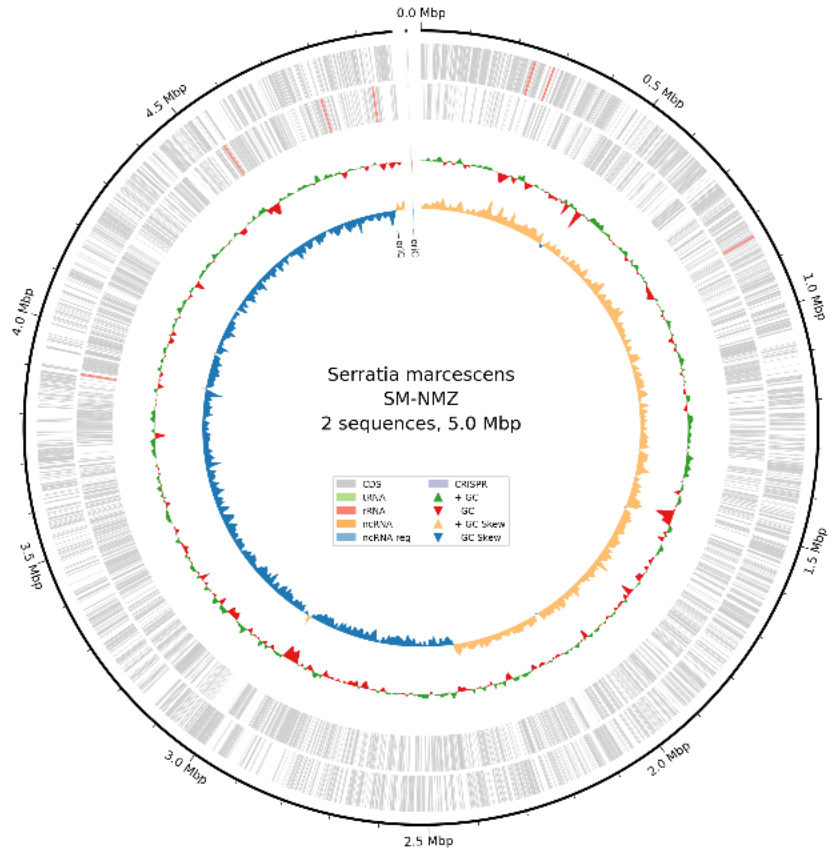
NMZ Serratia genomics

An explainer for the research group

SM-NMZ Slaughterhouse effluent
C14-NMZ Calotropis rhizosphere



6 November 2026

DNA, gene predictions and activity

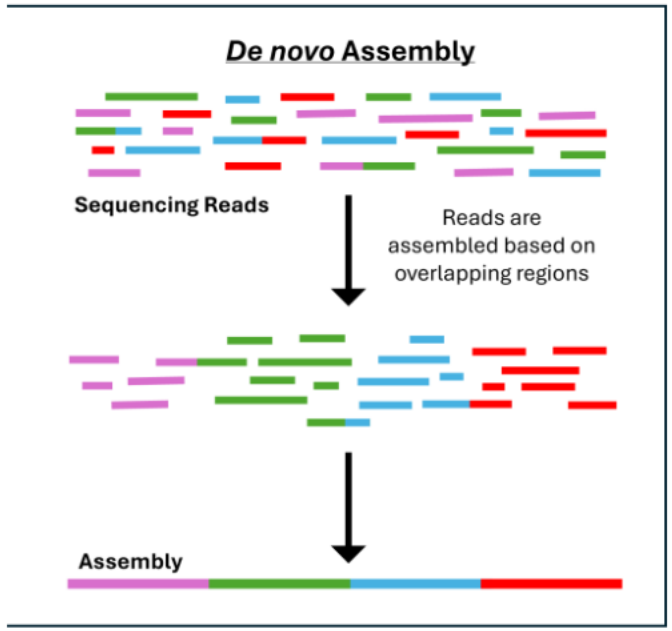
Stage	What it supplies
Sequencing	DNA fragments called reads
Assembly	Reconstructed chromosome and other replicons
Annotation	Predicted genes and possible functions
Experiment	An observed effect under stated conditions

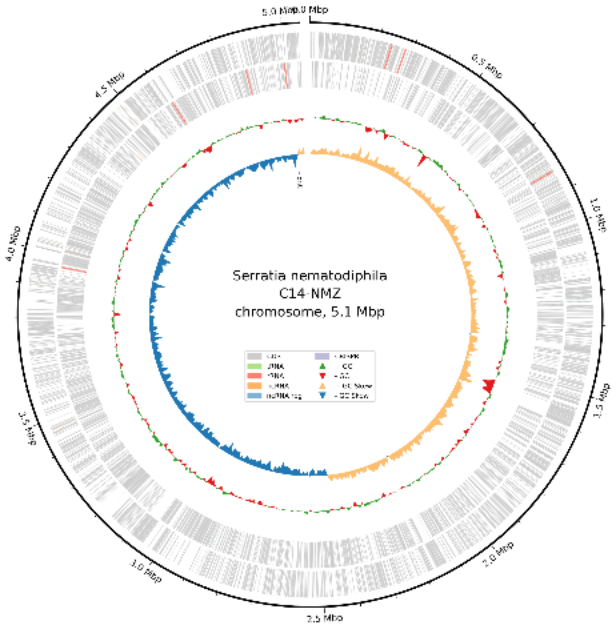
Gene predictions give us candidates to compare with lab assays.

Reads, assembly and genome annotation

Basecalled reads → Draft assembly → Annotated genome

```
@SRR014849.1 EIXKN4201CFU84 length=93
GGGGGGGGGGGGGGGGCTTTTTTTGTTTGGAACCGAAAGG
GTTTTGAATTTCAAACCCTTTTCGGTTTCCAACCTTCCAA
AGCAATGCCAATA
+SRR014849.1 EIXKN4201CFU84 length=93
3+&$#""""""""""""7F@71,'";C?,B;?6B;:EA1EA
1EA5'9B:?:#9EA0D@2EA5':>5?:%A;A8A;?9B;D@
/=<?7=9<2A8==
```





DNA bases + quality

Overlaps → contigs

Features + predictions

EMBL-EBI examples (left, centre) · C14 Bakta map (right): GTDB S. nematodiphila (slide 5)

What each kind of evidence answers

Evidence	Question it addresses
Related gene or protein	Is there a plausible candidate?
Several linked genes	Does the proposed machinery fit together?
Chemical analysis	Which molecular assignment has support?
Activity assay	What effect occurred in this experiment?
Controlled perturbation	Does this factor contribute to that effect?

Genome resources and identity

Feature	SM-NMZ	C14-NMZ
Chromosome (closed circular)	4.971 Mb	5.051 Mb
Additional replicon	3.581-kb circular plasmid	None recovered
Bakta CDSs	4,566	4,626
GTDB r232 label	S. marcescens	S. nematodiphila
TYGS species / subspecies cluster	6 / 11	6 / 10

Closed assemblies let us compare 35–44 kb neighborhoods without contig breaks.

TYGS groups both as S. marcescens. The Bakta and pig plots retain C14's GTDB label.

ANI and digital DNA–DNA hybridization

Measure	What is compared	How it is interpreted
ANI + aligned fraction	DNA sequence identity in aligned regions	Identity together with alignment coverage
dDDH	Sequence-based estimate of DNA–DNA relatedness	Estimate and confidence interval

ANI = average nucleotide identity; AF = aligned fraction

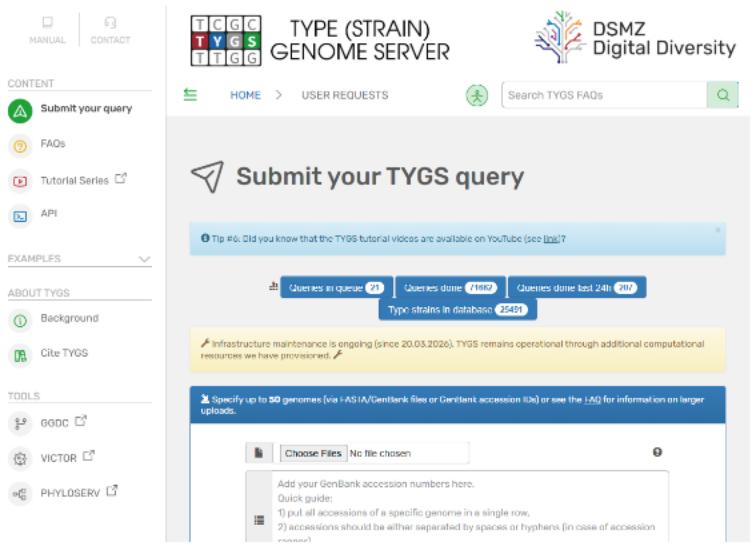
Broad species guides: ANI ≈95–96%; dDDH ≈70%

GTDB applies a representative-specific ANI radius.

NMZ pair: dDDH(d4) 74.2% (71.2–77.0%)

GTDB-Tk and TYGS workflows

TYGS · web service



Genome FASTA → type-strain comparisons
GBDP → dDDH + genome phylogram

GTDB-Tk · local pipeline

GTDB = Genome Taxonomy Database

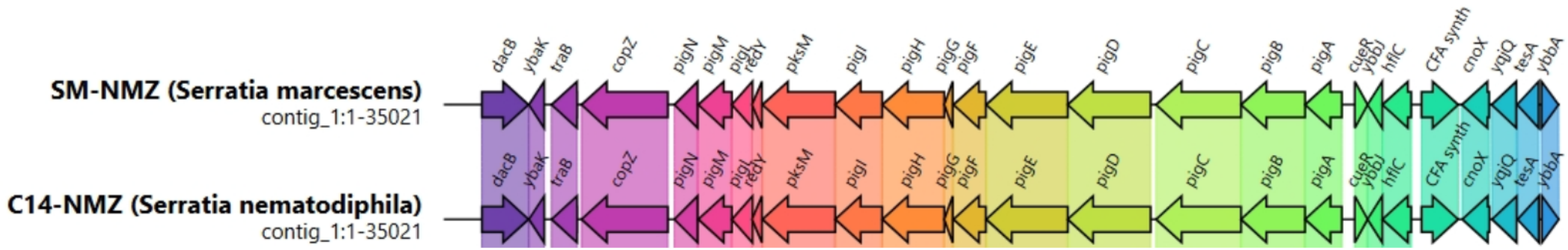
Genome FASTA + GTDB references
↓
ANI + aligned fraction
↓
Representative-specific species boundary
↓
Classification summary

NMZ queries were classified by ANI screening.

C14: GTDB S. nematodiphila; TYGS S. marcescens

Genes, neighborhoods, clusters and operons

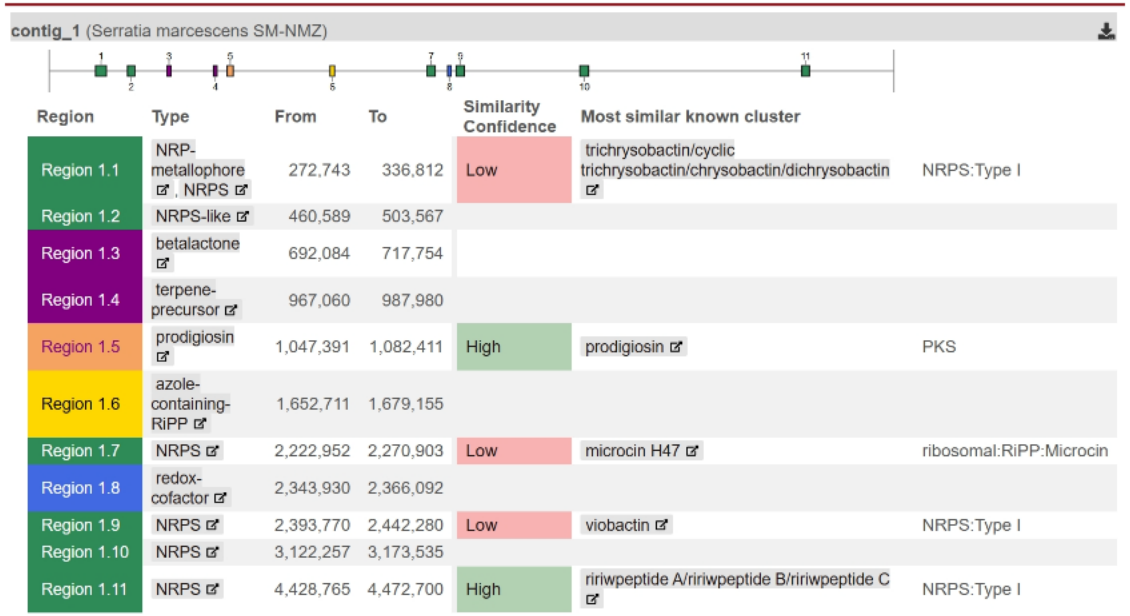
Term	Meaning
Gene	DNA sequence encoding a product
Neighborhood	Nearby genes: physical context
Biosynthetic cluster	Nearby genes implicated in making a molecule
Operon	Genes transcribed together as a unit



pig: a multi-gene example. SwrW: a multifunctional enzyme encoded by one gene.

Genome-wide BGC mining with antiSMASH

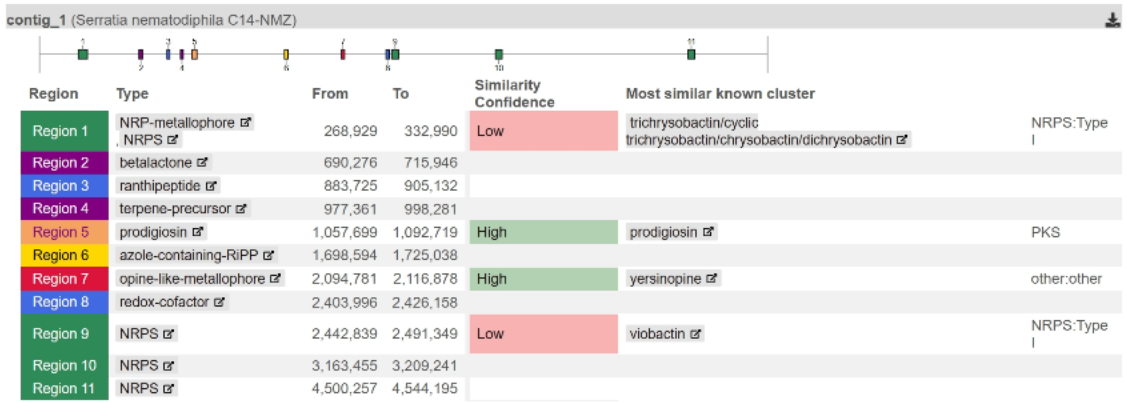
SM-NMZ · 11 predicted regions



contig_1 (Serratia marcescens SM-NMZ)

Region	Type	From	To	Similarity Confidence	Most similar known cluster	
Region 1.1	NRP-metallophore, NRPS	272,743	336,812	Low	trichrysobactin/cyclic trichrysobactin/chrysobactin/dichrysobactin	NRPS:Type I
Region 1.2	NRPS-like	460,589	503,567			
Region 1.3	betalactone	692,084	717,754			
Region 1.4	terpene-precursor	967,060	987,980			
Region 1.5	prodigiosin	1,047,391	1,082,411	High	prodigiosin	PKS
Region 1.6	azole-containing-RiPP	1,652,711	1,679,155			
Region 1.7	NRPS	2,222,952	2,270,903	Low	microcin H47	ribosomal:RiPP:Microcin
Region 1.8	redox-cofactor	2,343,930	2,366,092			
Region 1.9	NRPS	2,393,770	2,442,280	Low	viobactin	NRPS:Type I
Region 1.10	NRPS	3,122,257	3,173,535			
Region 1.11	NRPS	4,428,765	4,472,700	High	ririwpeptide A/ririwpeptide B/ririwpeptide C	NRPS:Type I

C14-NMZ · 11 predicted regions



contig_1 (Serratia nematodiphila C14-NMZ)

Region	Type	From	To	Similarity Confidence	Most similar known cluster	
Region 1	NRP-metallophore, NRPS	268,929	332,990	Low	trichrysobactin/cyclic trichrysobactin/chrysobactin/dichrysobactin	NRPS:Type I
Region 2	betalactone	690,276	715,946			
Region 3	ranthipeptide	883,725	905,132			
Region 4	terpene-precursor	977,361	998,281			
Region 5	prodigiosin	1,057,699	1,092,719	High	prodigiosin	PKS
Region 6	azole-containing-RiPP	1,698,594	1,725,038			
Region 7	opine-like-metallophore	2,094,781	2,116,878	High	yersinopine	other:other
Region 8	redox-cofactor	2,403,996	2,426,158			
Region 9	NRPS	2,442,839	2,491,349	Low	viobactin	NRPS:Type I
Region 10	NRPS	3,163,455	3,209,241			
Region 11	NRPS	4,500,257	4,544,195			

antiSMASH v8.0.4 lists regions in chromosome order. Check the underlying matches before using the displayed product name.

Shared regions include pig (5), the PQQ cofactor cassette (8) and swrW (11).

C14 adds ranthipeptide (3) and opine-like (7) calls. Region 10 differs between isolates.

Reading antiSMASH reference matches

Region	Match or architecture	Interpretation
5 · pig	16 proteins, 87–100% identity; near-full-length matches	Conserved 14-gene prodigiosin pathway
11 · swrW	SM “ririwpeptide” hit: 1 protein, 47% identity, 45.9% coverage	Curated SwrW call fits the Ser-selective module and reported chemistry
1, 9 and C14 7 · metal binding	Partial MIBiG matches, 5–6 proteins	Supports metallophore class potential
10 · focal NRPSs	SM: Gap1-like, 3 modules C14: 1 module, predicted D-Val	Neither architecture supports a stephensiolide G/K assignment.

Compare how much matches, which genes match and how their domains are arranged.

SwrW evidence in the NMZ study

Observation or test	What it supports
SwrW sequence, domains and context	A plausible biosynthetic candidate
Putative LC-HRMS assignments	Serrawettin W1 (= serratamolide A) and serratamolide C
Tension or E24 assay	Effect of the particular material tested
Controlled intervention / purified compound	A test of contribution to that effect

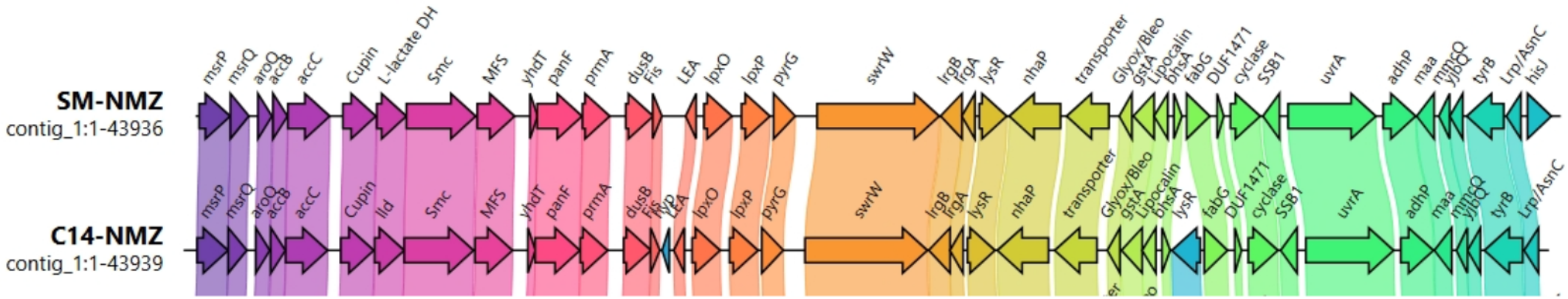
The conserved SwrW candidate and reported LC-HRMS assignments agree.

Pair the chemistry with activity measured on the same test material.

SwrW and reported serratamolides

Genomic evidence

Both isolates carry a conserved swrW candidate.



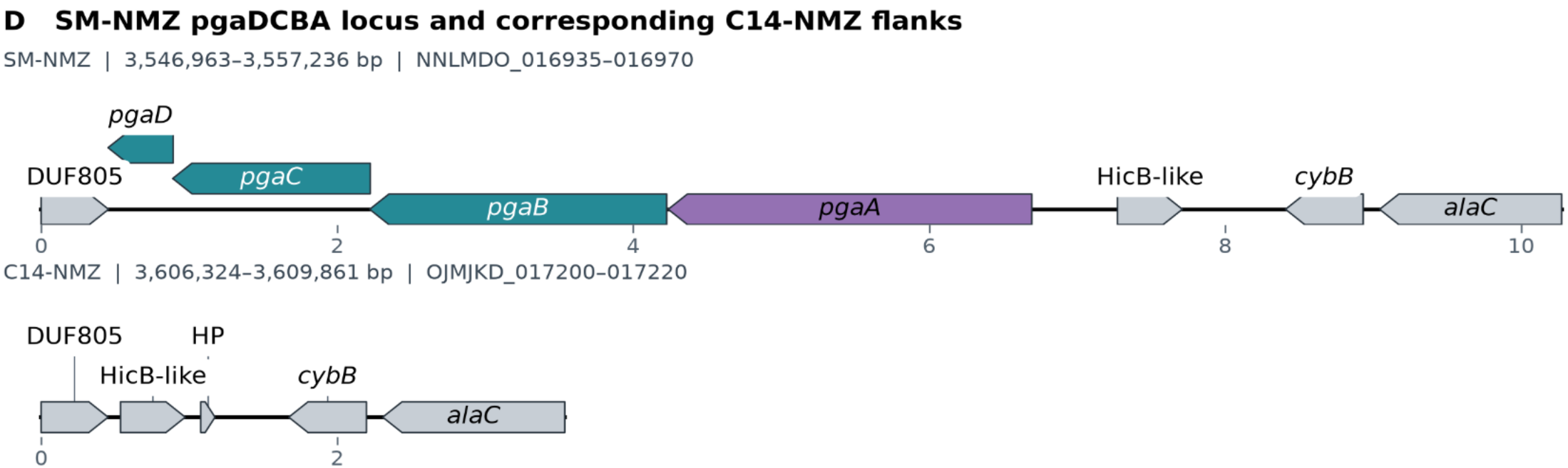
Putative LC-HRMS assignments · samples 1 and 2

Serrawettin W1 (= serratamolide A) and serratamolide C

The report does not map rows to isolates. Separately, pig fits the red/pink cultures.

Extracellular matrices and surface activity

Both isolates: capsule/EPS and cellulose candidates



Gene arrows. Scale in kb from the first displayed CDS.

SM-NMZ adds a contiguous pgaDCBA locus associated with PNAG.

Compare biofilm or emulsion stability if the isolates differ in those assays.

EPS, CPS and matrix polysaccharides

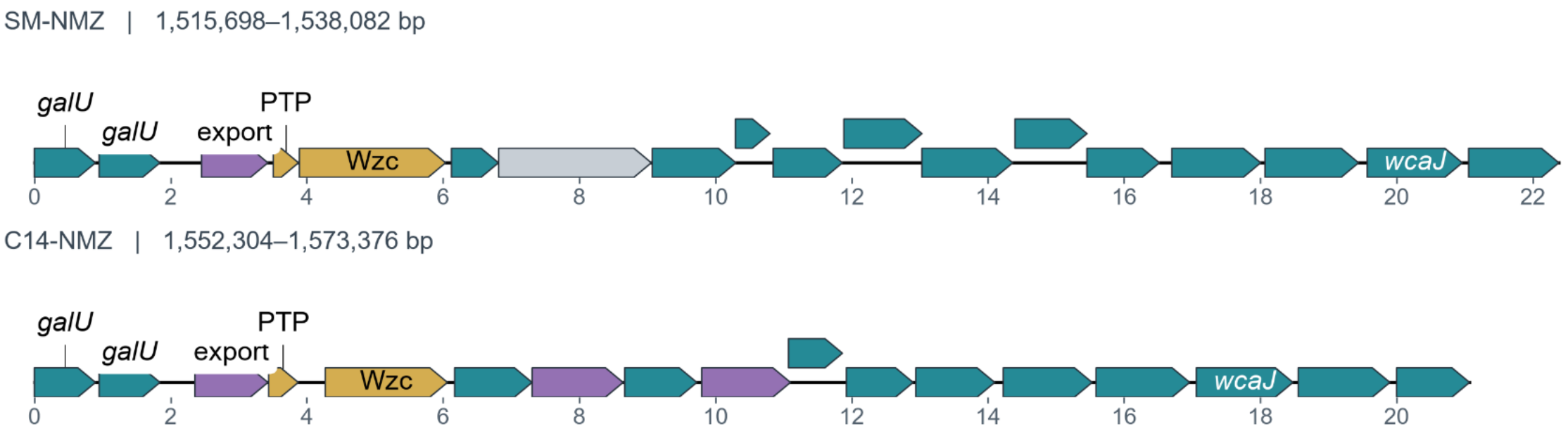
Polysaccharides are long chains of sugar units.

Term	What it means
EPS	Extracellular polymeric substances: the matrix
CPS	Capsular polysaccharide: associated with the cell
PNAG	β-(1→6)-linked N-acetylglucosamine polymer
Cellulose	β-(1→4)-linked glucose polymer

Small lipopeptides and matrix polymers can affect emulsions differently.

Record whether each assay used whole broth, supernatant or an extract.

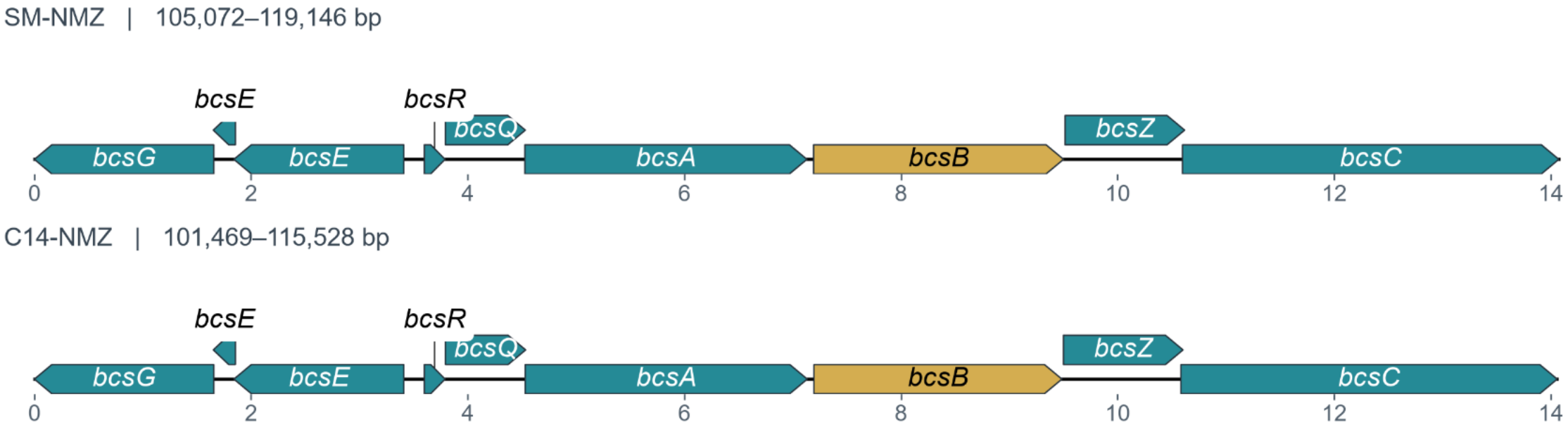
Capsule loci and the HADEG markers



Wzb-like PTP and Wzc are capsule-associated phosphatase and kinase candidates.

The HADEG “Emulsan” hits sit in capsule-associated neighborhoods.

Cellulose-associated loci in both isolates

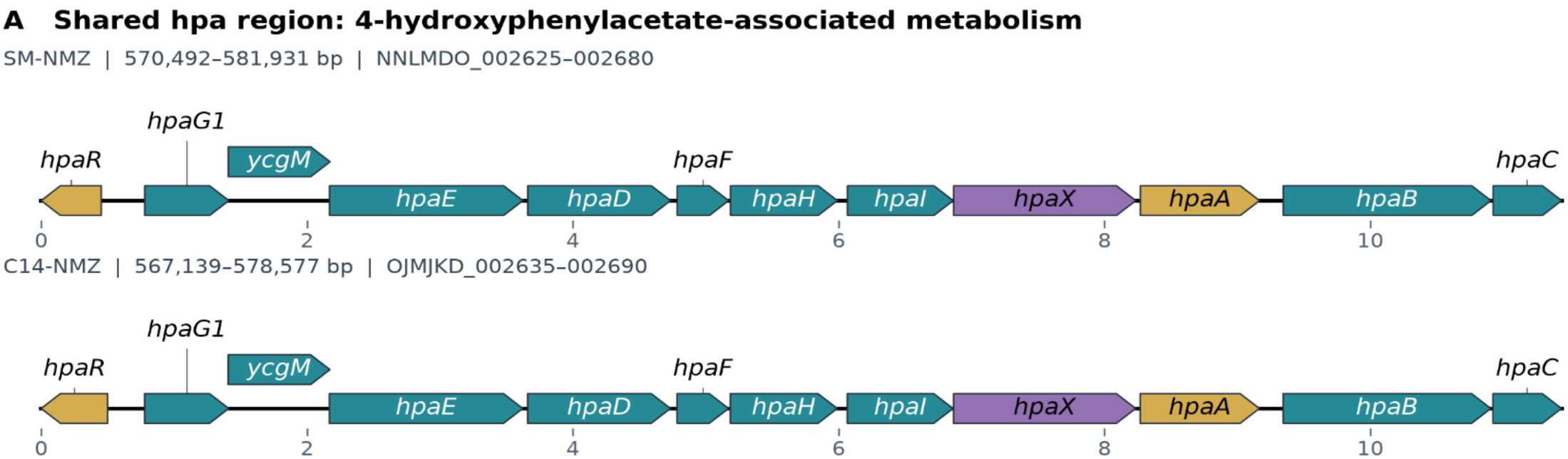


BcsA: synthesis · BcsB: partner · BcsC: export-associated

Scale: kb. Conserved gene order supports shared biosynthetic potential.

Aromatic metabolism depends on the substrate

Both isolates: organized hpa and paa loci



Gene arrows. Scale in kb from the first displayed CDS.

SM adds hydroxybenzoate/pca and gentisate candidates.

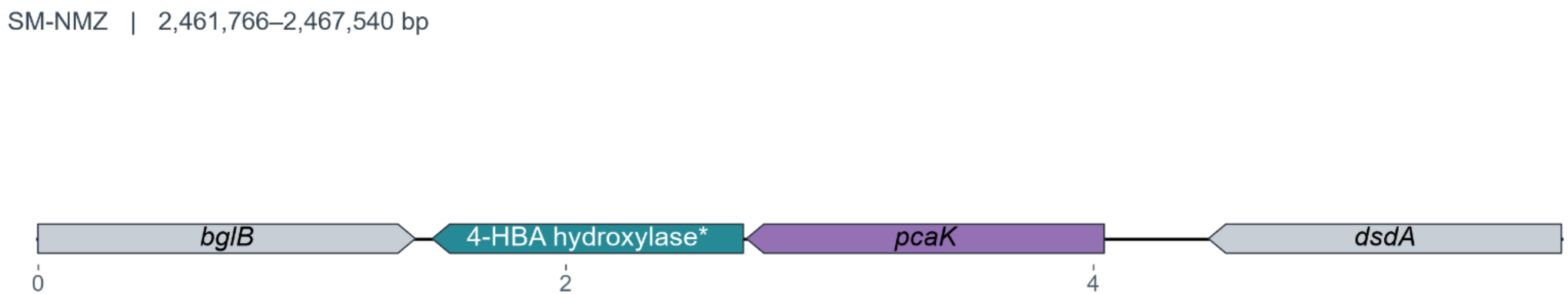
Match conversion assays to the oxygenated aromatic substrate and its route.

The HADEG Toluene_degradation hit

A characterized reference route · P. mendocina KR1

toluene → p-cresol → 4-HBA → protocatechuate

Initial activation and processing precede the PobA step.



SM NNLMDO_011765 and pcaK point to a downstream aromatic-acid step.

4-HBA = 4-hydroxybenzoate. *Label abbreviates the source product.

LadA and the alkane-oxidation hypothesis

Historically proposed role

Long-chain alkane → primary alcohol

LadA was reported as a soluble, flavin-dependent candidate for this first step.

Why search here?

To identify a plausible initiating enzyme alongside downstream carbon metabolism.

LadA evidence across the literature

Study	Evidence reported	Interpretation
Feng et al. 2007	Canonical LadA: in-vivo and in-vitro support	Initial long-chain oxidation proposed
Boonmak et al. 2014	B23 homologs: alkane depletion in a heterologous host	Cell-based evidence; not purified enzyme
Mireles et al. 2025	No direct hydroxylation detected for tested proteins	First-step assignment questioned

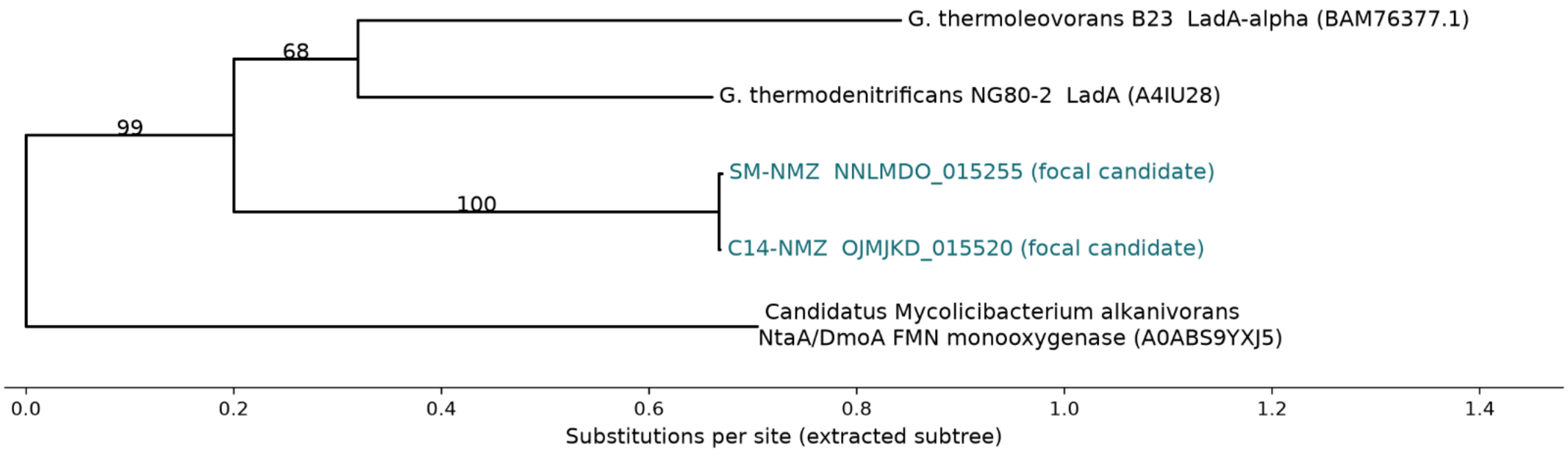
The literature gives us a candidate family and a substrate question to test.

Initial alkane oxidation

Simplified step	Genomic interpretation
Alkane to alcohol	Initiating system unresolved
Alcohol to aldehyde	Downstream processing candidates
Aldehyde to fatty acid	Downstream processing candidates
Fatty acid to cellular metabolism	Beta-oxidation and assimilation context

Both genomes encode β-oxidation. Alkane initiation remains unresolved.

The LadA-like comparison



Node labels show UFBoot support (%).

UFBoot support: 100% for the focal pair, 99% for the LadA-associated clade.

The conserved ssuD and metNPQ context adds a sulfur-related substrate hypothesis.

Matching assays to genomic candidates

Endpoint, if measured	Relevant interpretation	Evidence to pair with it
Surface/interfacial tension	Amphiphile contribution	Material, interface, mN/m, control
E24 emulsification index	Amphiphile and/or matrix contribution	Oil, 24-h protocol, layer heights
Aromatic conversion	Route matching the exact substrate	Substrate, products and recovery controls
Pour point	Low-temperature oil-flow performance	Oil, dose, method and temperature

Pair each assay value with the material tested and its control.

Host-interaction genes

Reference-associated system	Biological role in characterized systems
ShlA–ShlB	Hemolysin machinery
Has	Heme acquisition
Type VI secretion system	Bacterial competition

These homologs provide context for host and microbial interactions.

Evidence scoreboard

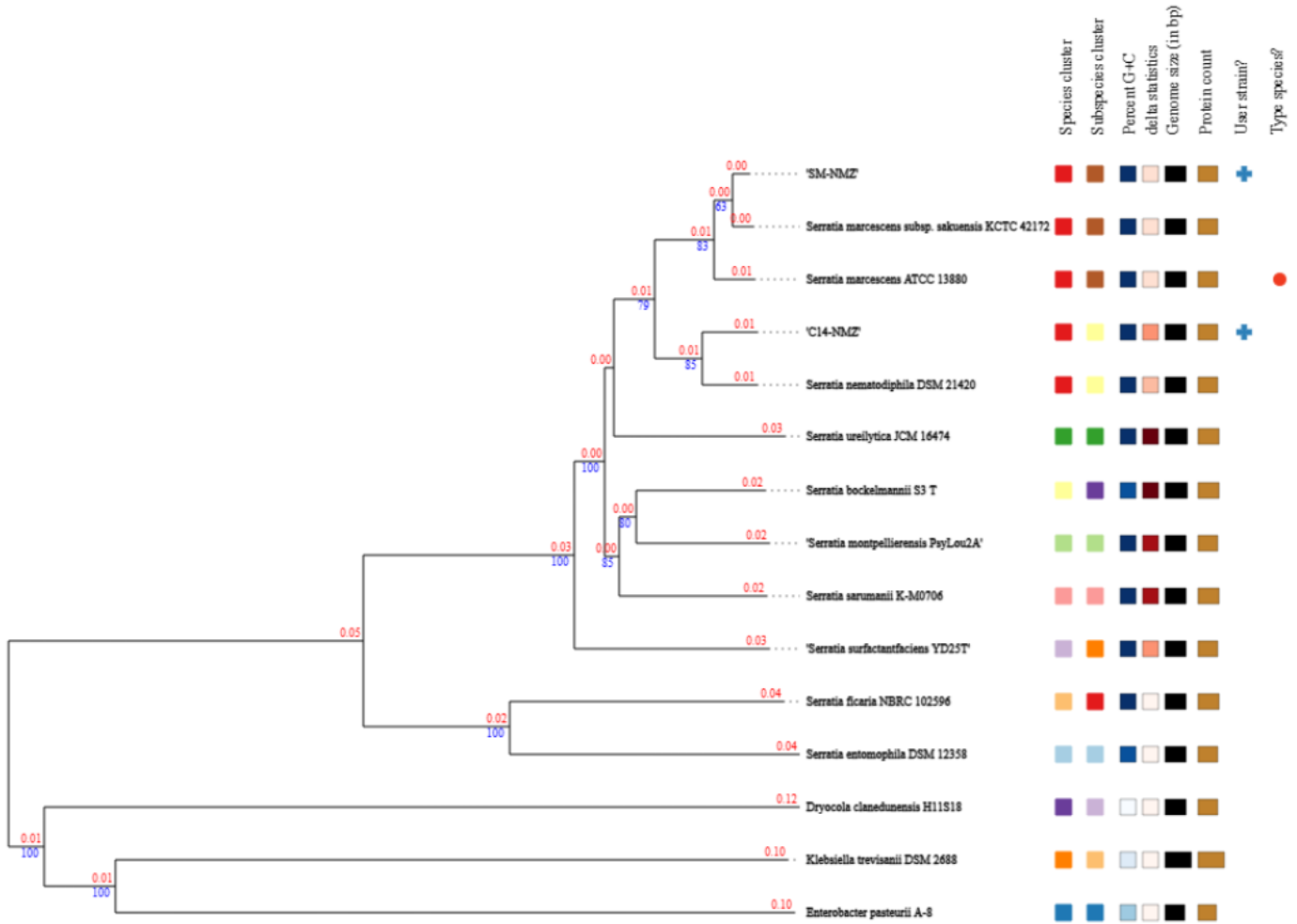
System / locus	Genomic candidate	Chemistry / phenotype in hand	Wet-lab assay slot
Biosurfactant (swrW)	Shared 1-module NRPS 98.9% protein identity	Putative LC-HRMS: serratamolide A (W1) and C	Tension, E24 and pour point
Pigment (pigA–pigN)	Conserved 14-gene pathway	Red/pink cultures	Pigment identity, if measured
Matrix polysaccharides	Shared CPS/EPS and bcs; SM pgaDCBA	Polymer chemistry unmeasured	E24 or biofilm assays
Oxygenated aromatics	Shared hpa/paa; SM hydroxybenzoate/pca/gentisate	Conversion data pending	Substrate-specific conversion
Alkanes	β-oxidation; LadA-like candidate	Initial oxidation unresolved	Alkane conversion and products

Add the measured endpoints and test materials to the matching rows.

Workflow and tools by purpose

Question	Tools and evidence
Which read evidence enters assembly?	NanoPlot, Chopper, Filtlong
How are sequences reconstructed?	Autocycler/assemblers, dnaapler, Dorado
Where are genes and what are candidates?	Bakta, antiSMASH, DRAM, CANT-HYD, HADEG
How are genomes and proteins related?	TYGS, GTDB-Tk, ProteinOrtho/PoFF, IQ-TREE
How are local arrangements compared?	clinker and DNA Features Viewer

The TYGS genome phylogram



Blue labels: pseudo-bootstrap support (%) from 100 replicates.

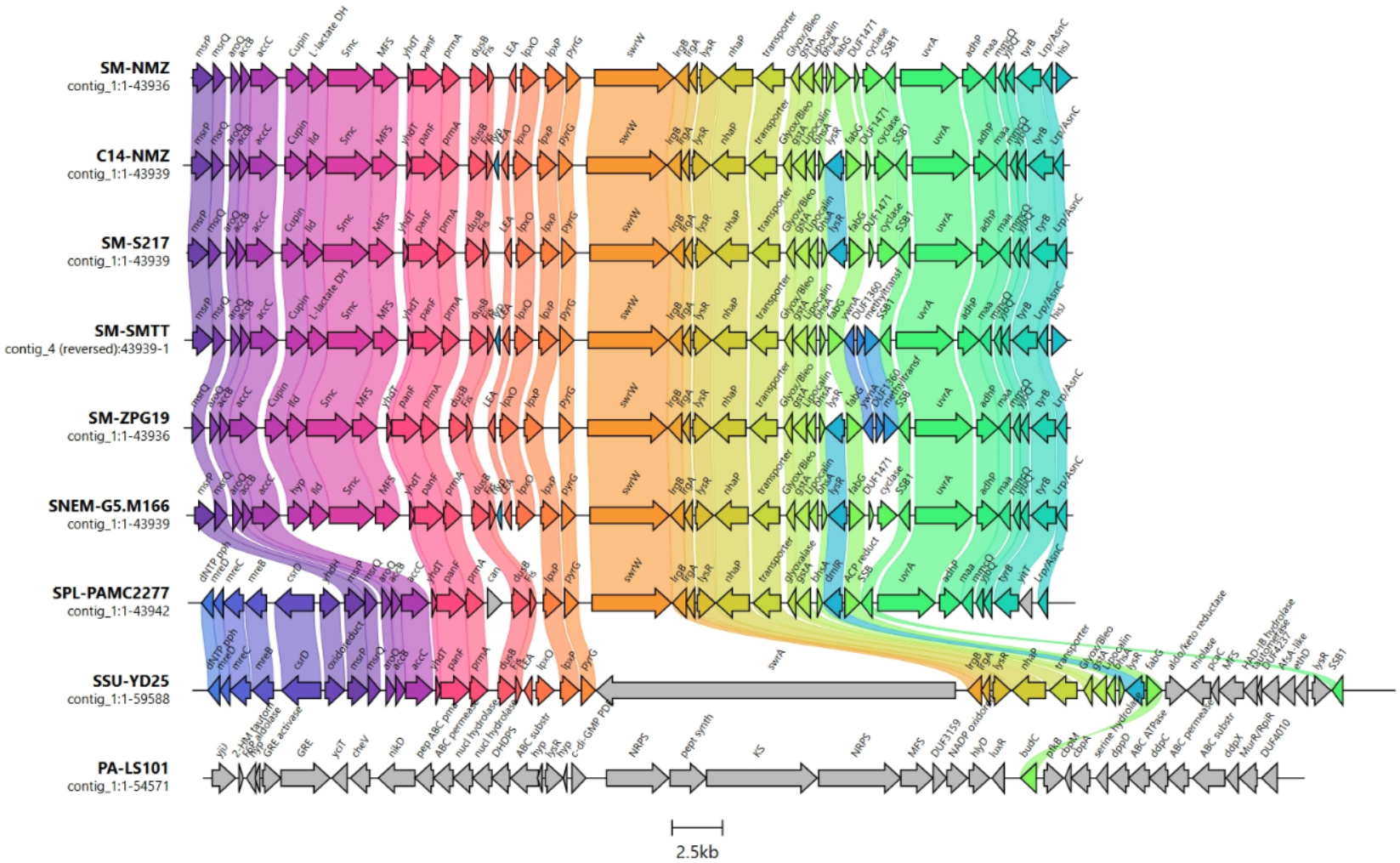
Taxonomic comparisons

dDDH(d4), %	SM-NMZ	C14-NMZ
S. marcescens ATCC 13880	90.7	73.7
S. nematodiphila DSM 21420	74.7	85.5
Direct focal comparison	74.2 (71.2–77.0)	Same comparison

GTDB ANI screening

SM 98.86% ANI (AF 0.947)
C14 98.22% ANI (AF 0.932)

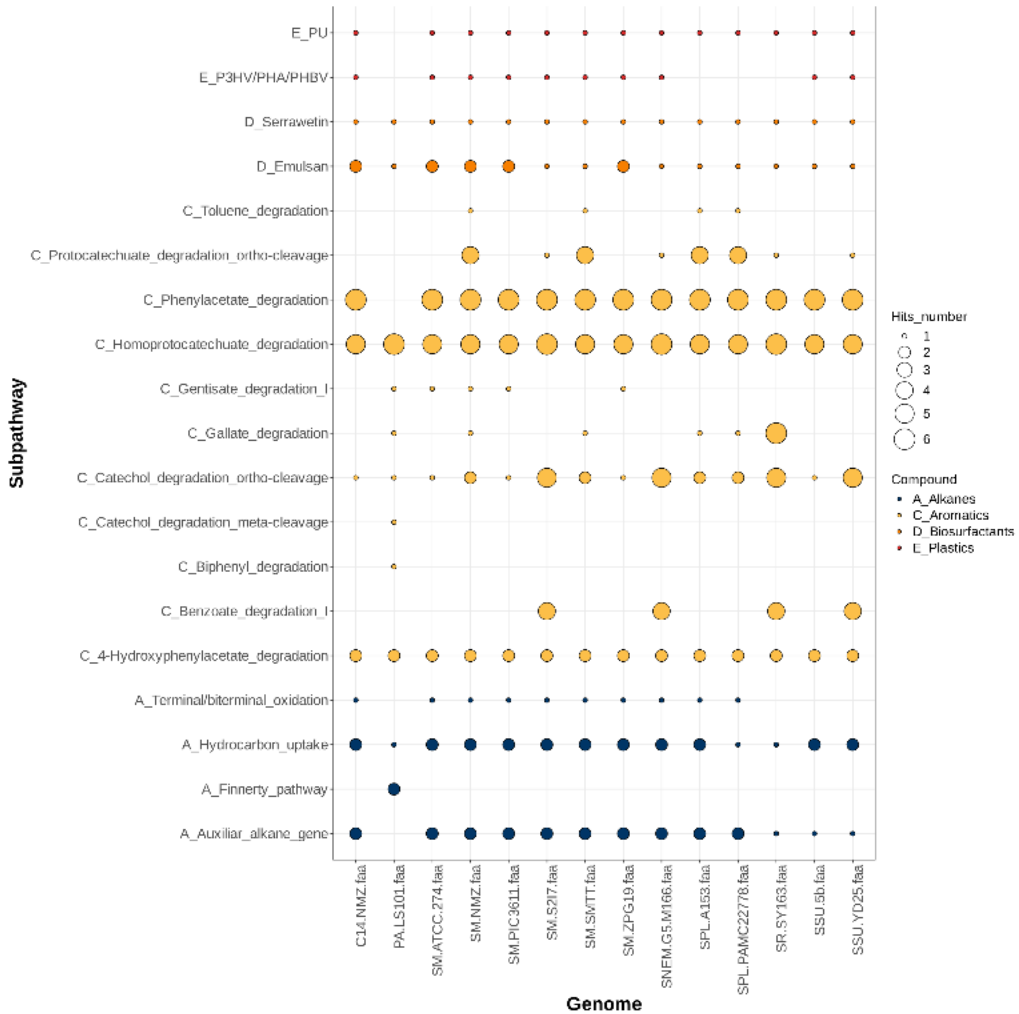
Reading the SwrW neighborhood



Arrows
Gene length and direction

Ribbons
Protein matches

Reading HADEG marker categories



Emulsan
Generic capsule markers

Toluene_degradation
Hydroxybenzoate-hydroxylase candidate

Bubble counts describe markers.
They do not measure pathway activity.

Chemical assignments and peak areas

Observation	What it can support
Exact mass and adduct	A candidate molecular composition
Fragmentation and reference match	Additional structural evidence
Authentic standard, when documented	A stronger identity comparison
Relative peak area	Detector signal under a stated denominator

Serrawettin W1 (= serratamolide A) and serratamolide C fit the SwrW candidate.

Additional SM aromatic candidates

C SM-NMZ hydroxybenzoate-hydroxylase / pcaK neighborhood

SM-NMZ | 2,461,766–2,467,540 bp | NNLMDO_011760–011775



D SM-NMZ protocatechuate-associated region

SM-NMZ | 3,121,817–3,130,237 bp | NNLMDO_014965–015005

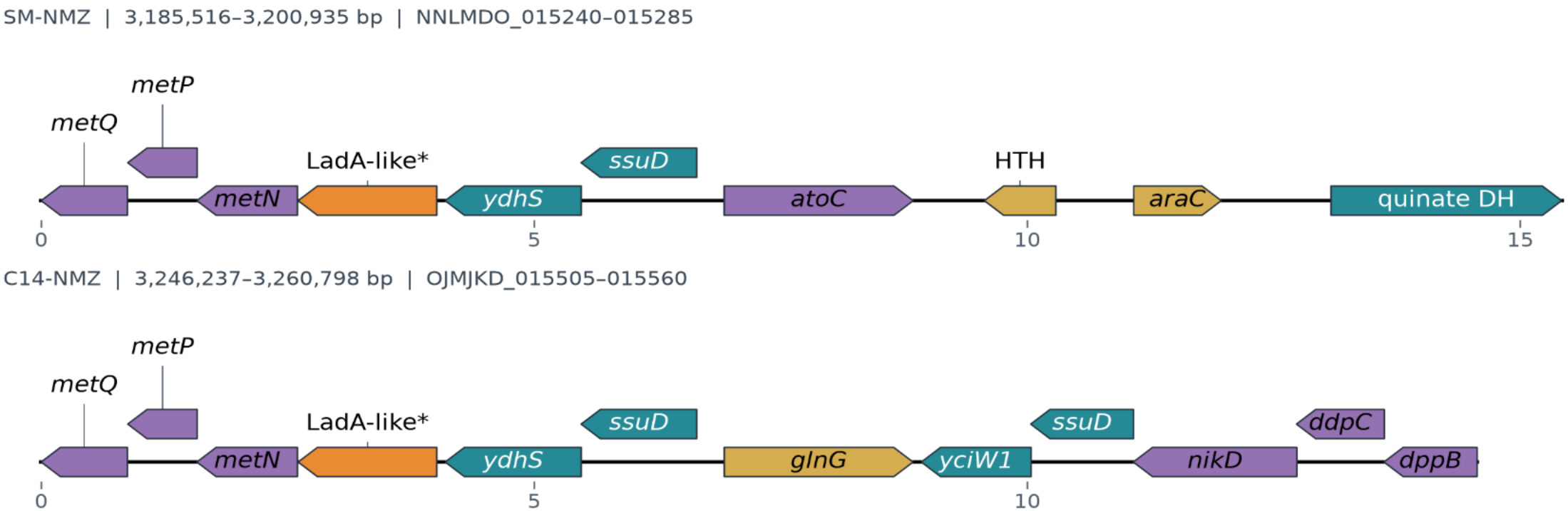


Scale in kb within each panel. Panels use different physical scales.

Additional candidates broaden the substrate hypotheses.

Some downstream steps remain unresolved.

LadA-like genomic neighborhoods

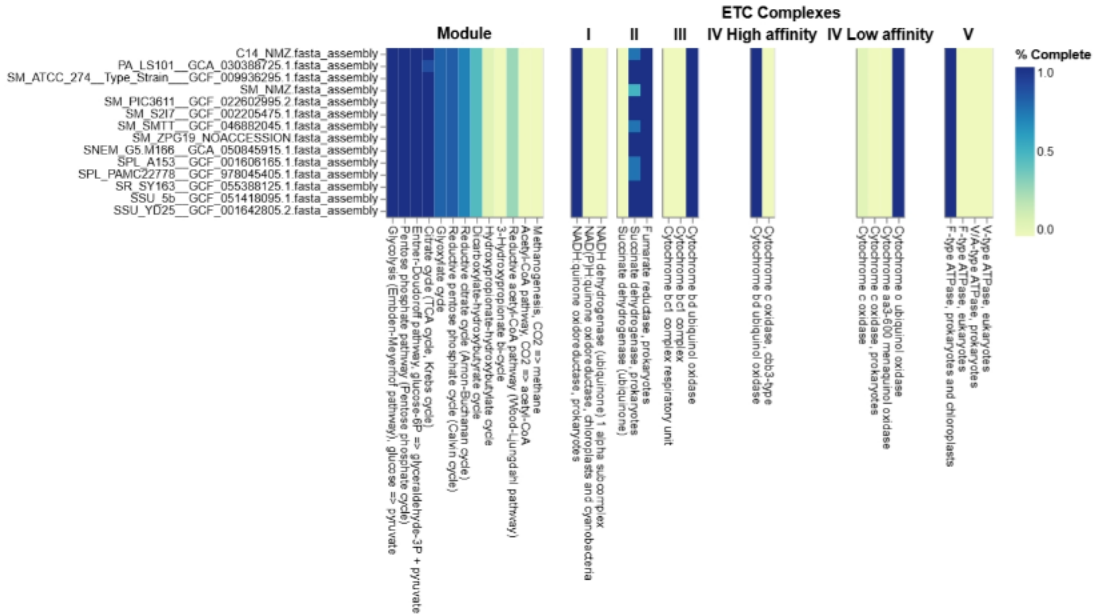


Gene arrows. Scale in kb from the first displayed CDS.

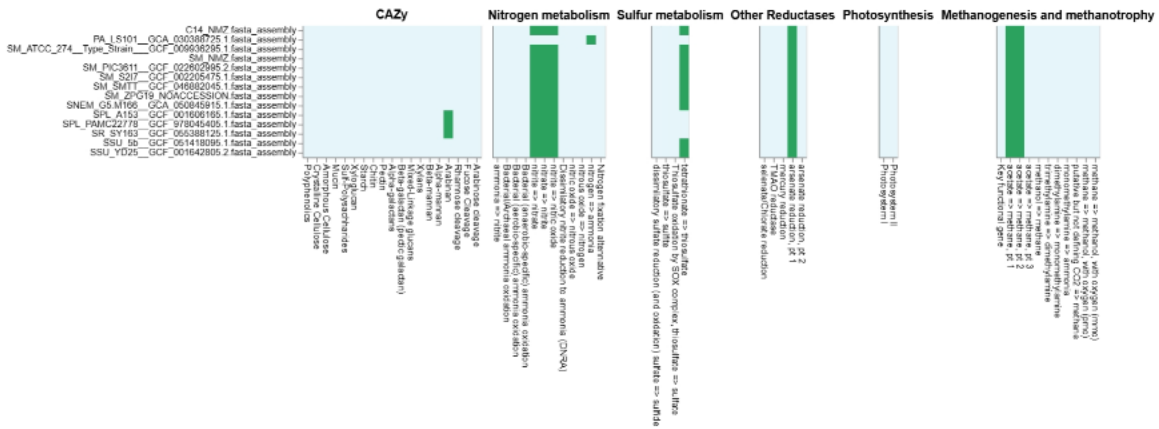
Source gene qualifier: ssuD. Candidate substrate remains unresolved.

Conserved metabolic background (DRAM)

Central carbon and respiration



Auxiliary functions



Within Serratia, three calls varied: succinate dehydrogenase, GH43 arabinan and a tetrathionate marker.

Invariant DRAM traits: 90/98 across all 14 genomes (91.8%).

Within Serratia: 95/98 across 13 genomes (96.9%), including shared absences.